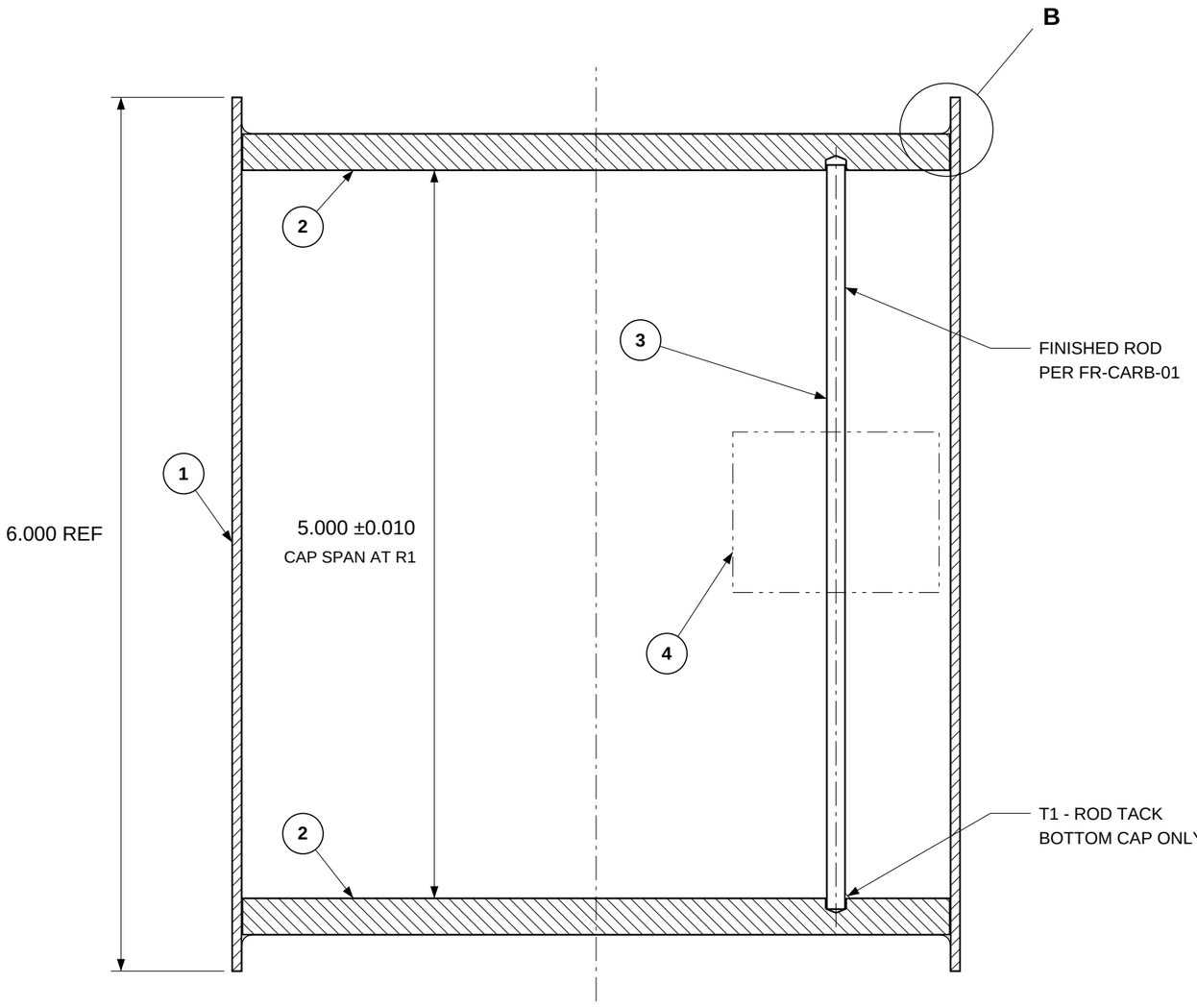
SECTION A-A

THROUGH VESSEL AND ROD AXES - SCALE 1:1

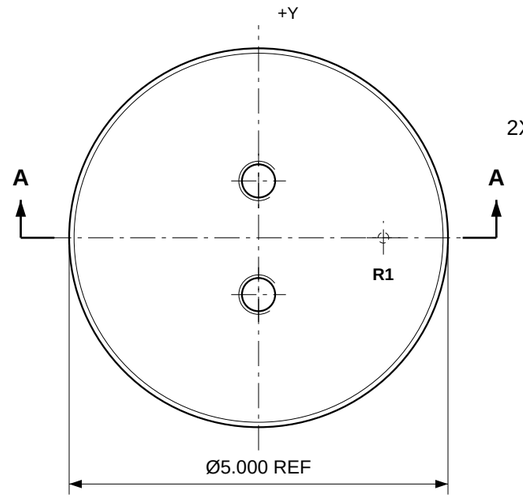


FIXED-LENGTH ROD SEATS IN BOTTOM REGISTER; TOP REGISTER RETAINS CLEARANCE.

PORTS LIE OUT OF THE SECTION PLANE. FLOAT ENVELOPE IS REFERENCE ONLY.

TOP OUTSIDE END VIEW

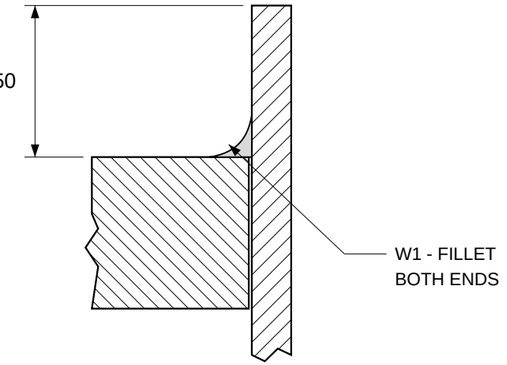
LOOKING DOWN -Z - SCALE 1:2



BOTH R1 REGISTERS SHARE +X; BOTH PORT PAIRS SHARE ±Y.

DETAIL B - CLOSURE JOINT

SCALE 4:1 - LOWER END MIRRORED



WELD PROFILE DIAGRAMMATIC; SIZE NOT ESTABLISHED.

CAP OD / BORE SLIP GAP SHOWN; DO NOT CHAMFER WELD ROOT.

PARTS AT CLOSURE

ITEM	QTY	MATERIAL	DESCRIPTION / DEFINITION
1	1	HSM-000008	316 SS TUBE; Ø5.000 x 0.065 WALL x 6.000
2	2	HSM-000009	316 SS FINISHED END CAP; END-CAP-01
3	1	HSM-000336	316 SS FLOAT GUIDE ROD; FR-CARB-01
4	1	HSM-000032	ASA AERO MAGNETIC FLOAT; INSTALL BEFORE CLOSURE

ASSEMBLY REQUIREMENTS

1. Fabrication state of HSM-000004 before permanent fittings. Component ports, registers and edge preparation: END-CAP-01. Stock dimensions are nominal.

2. Finished cap inside-face spacing AT ROD AXIS: 5.000 ±0.010. Set both outside faces 0.250 below adjacent tube rims. Align both port pairs on one diameter; both R1 register axes coaxial with the rod. Rod +X faces reservoir B; ports lie on ±Y in the cold core.

3. Use the fixed-length rod per FR-CARB-01. Seat and tack it in the bottom register before the bottom closure weld. Install the float before top closure; verify top engagement, free travel and full cap seating at the specified spacing.

4. W1: continuous internal-corner fillet around BOTH cap perimeters. ER316L .030 filler. Clean bore band and cap outside face; preserve the square cap weld root. T1: tack rod to bottom cap. Welding, inspection, hydro and passivation: pressure-vessel procedure, steps 2-8.

RELEASE ITEMS: W1 SIZE AND WELD QUALIFICATION; REMAINING UNSPECIFIED ASSEMBLY TOLERANCES.

HSM-000004 / FABRICATION STATE

DIMENSIONS GOVERN.

DO NOT SCALE.

PRINT AT 100%

FOR STATED SCALES.

REF = REFERENCE

HOME SODA MACHINE

CARBONATOR VESSEL - CLOSURE ASSEMBLY

PROTOTYPE

FOR REVIEW

MATERIAL	UNITS	QUANTITY
SEE PARTS LIST	**INCHES**	**1 PER MACHINE**

DRAWING	REV	DATE	SCALE	SHEET
CV-WLD-01	**B**	**2026-10-26**	**AS SHOWN**	**1 OF 1**